



Assignment 1

Objective:

- Use a timer to control a servo motor to move in 10-degree increments every second, from 0 to 180 degrees, and then return to 0 degrees.

Components:

- ATmega32 microcontroller
- Servo motor
- Resistors
- wires

Circuit Setup:

- Connect the servo motor to a PWM output pin on the ATmega32.
- Ensure the power supply matches the servo motor's requirements.

Programming:

- Initialize the timer to generate a 1-second delay.
- Configure a PWM signal to control the servo motor's position.
- Increase the servo's angle by 10 degrees every second using PWM.
- Once 180 degrees is reached, reset the position back to 0 degrees.

Operation :

- The servo should move in increments of 10 degrees every second until it reaches 180 degrees, then return to the starting position and repeat.